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EXTENDED ABSTRACT

Hybrid dynamical-statistical modelling of the North Atlantic climate

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1. Introduction and objectives

Decadal climate prediction, which aims to forecast climate up to ten years ahead, has the potential to inform decision-making across many sectors. On this timescale, numerical climate models demonstrate skill in predicting sea surface temperature (SST), particularly in the North Atlantic, due to the slow evolution of ocean variability. While several climate studies have demonstrated that SST anomalies can influence atmospheric circulation and the occurrence of specific weather patterns, predictions of impactful climate anomalies show limited skill over continental Europe. The limited skill of models has been suggested to be linked to a poor simulation of the atmospheric response to SST anomalies (Scaife and Smith, 2018). To address this gap, we propose a statistical correction framework for the North Atlantic-European (NAE) climate predictions using North Atlantic SST from dynamical models as the sole predictor.

2. Methods and data

Sea-level pressure (SLP), total precipitation (TP), and 2-meter air temperature over the continents (T2m) in the NAE sector (defined as 20°N–70°N, 100°W–40°E) are obtained from the ERA5 Monthly Data Reanalysis (Hersbach et al., 2023) for the period 1940–2025. Monthly SST in the 20°S–70°N, 70°W–20°E domain from the Community Earth System Model (CESM) decadal prediction experiments (Yeager et al., 2018) is used to build the hybrid forecasts. These decadal experiments are initialized with observed conditions on November 1 of each start year spanning 1960–2017, with an ensemble size of 40 members, and run for ten lead years. For each start year, the ensemble-mean SST is computed, and anomalies are calculated relative to the climatological mean over all start years.

Lead-time averages are then calculated over years LT_{start}–LT_{end} after initialization, providing a smoothed $(LT_{\mathrm{end}} - LT_{\mathrm{start}} + 1)$-year forecast window—for example, LT3–7 corresponds to averaging years 3–7 after initialization, so a start year of 1960 corresponds to the mean SST over 1963–1967.

Figure 1 provides an overview of the proposed methodological pipeline. First, a Variational Autoencoder (VAE) is trained to reduce the dimensions of the NAE SLP, TP, and T2m fields. The VAE is implemented as a fully connected encoder–decoder with two hidden layers (512 and 258 neurons) and ReLU activations, mapping the three large-scale ERA5 atmospheric fields into a Gaussian latent vector

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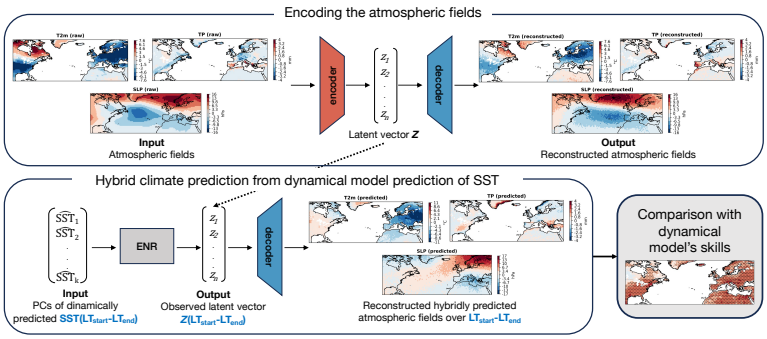

Figure 1. *Schematic that visually summarizes the research.*

Z of dimension n. The optimal n is selected by jointly considering the variance explained and the contribution of each latent dimension to the Kullback–Leibler divergence. Winter (DJF) and summer (JJA) months are treated separately to account for seasonal differences.

Second, we fit an elastic net regression (ENR) model relating the first k principal components (PCs) of CESM-predicted North Atlantic SSTs ($\widehat{\mathrm{SST}}$), averaged over lead-times LT_{start}–LT_{end}, to the ERA5-derived latent vector Z, which is averaged over the corresponding years to match the lead-time window of the SST forecasts. The ENR combines ℓ_1 and ℓ_2 regularization, promoting sparsity while handling collinearity among the PCs. The predicted latent vector Z is decoded through the VAE to reconstruct the SLP, TP, and T2m fields, enabling evaluation of prediction skill. These skills are compared against dynamical decadal prediction models. Leave-5-years-out cross-validation is applied to assess generalizability while accounting for the temporal autocorrelation of the data.

3. Preliminary results and Expected outcomes

Preliminary results demonstrate that the VAE effectively reduces the dimensionality of NAE climate fields. For winter and summer, optimal latent dimensions of $n = 6$ and $n = 7$ capture approximately 70% of the variance, respectively, while preserving dominant large-scale circulation patterns. k-means clustering further confirms the physical interpretability of the latent space, identifying the principal circulation regimes in winter ($k = 4$) and summer ($k = 5$).

The framework using $k = 15$ SST PCs as input demonstrates significant skill in predicting the North Atlantic Oscillation at lead times of 3-7 years, with a correlation coefficient of 0.5 and an improved variance relative to raw dynamical forecasts. Significant skill beyond the linear trend is found for winter precipitation and T2m over central western Europe, while skill in summer is low to non-significant.

An analysis of the mechanisms underlying predictability is underway, including an examination of skillful SST patterns and associated climate anomalies over land. We also plan to assess the predictive skill of the method with respect to external forcing and internal variability. Potential extensions include using SST data from multi-model ensembles beyond CESM, as well as comparing the linear model with nonlinear approaches to evaluate whether more complex SST–atmosphere relationships can be captured. These results suggest that statistically correcting the atmospheric response to dynamically predicted North Atlantic SSTs can meaningfully enhance decadal predictability over the NAE sector.

References

Scaife, A. A., & Smith, D. (2018). A signal-to-noise paradox in climate science. *npj Climate and Atmospheric Science* **1**, 28.

Hersbach, H., et al. (2023). ERA5 Monthly Averaged Data on Single Levels from 1940 to Present. *Copernicus Climate Change Service (C3S) Climate Data Store (CDS).*

Yeager, S. G., et al. (2018). Predicting near-term changes in the earth system: A large ensemble of initialized decadal prediction simulations using the community earth system model. *Bulletin of the American Meteorological Society* **99**, 1867–1886.